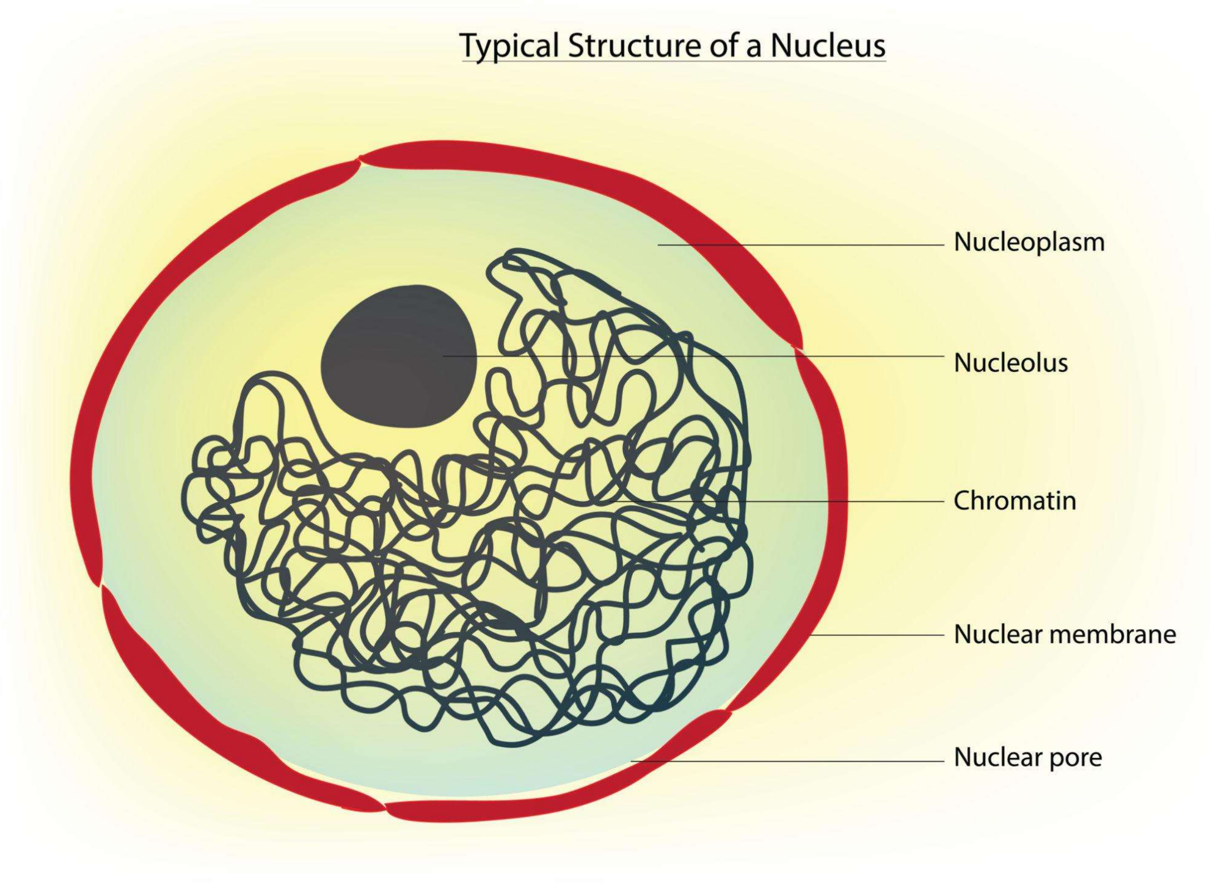
Structure and Function of the Nucleus

Labeled Diagram of Nucleus



INTRODUCTION

The **nucleus** is a **prominent, membrane-bound organelle** found in all eukaryotic cells. It is usually spherical and centrally located, though position may vary. It was first discovered by **Robert Brown (1831)**.

It acts as the **control center** of the cell because it:

- Stores hereditary material (**DNA**)
- Regulates gene expression
- Coordinates cell division and metabolism

ULTRASTRUCTURE OF NUCLEUS (DETAILED)

The nucleus is highly organized and consists of the following components

1. Nuclear Envelope (Karyotheca)

- A **double-layered membrane** enclosing the nucleus
- Each membrane is a **phospholipid bilayer**
- Thickness ≈ **10–20 nm**
- The space between the two membranes is called the **perinuclear space**

Special Features:

- Outer membrane is continuous with **endoplasmic reticulum (ER)**
- Inner membrane is supported by a protein layer called the **nuclear lamina**

Functions:

- Separates nucleoplasm from cytoplasm
- Maintains nuclear shape
- Provides mechanical support
- Regulates exchange via pores

2. Nuclear Pore Complex (NPC)

- Numerous pores (≈ **3000–4000 per nucleus**)
- Each pore is made of complex proteins forming a **nuclear pore complex**

Transport Mechanism:

- **Passive transport**: small molecules
- **Active transport**: proteins, RNA (requires energy)

Functions:

- Controls selective exchange of materials
- Allows:
    - mRNA → cytoplasm
    - Ribosomal subunits → cytoplasm

    - Proteins → nucleus

3.Nucleoplasm(Karyolymph)

Semi-fluid, dense matrix inside nucleus Contains enzymes, nucleotides, ions, chromatin

Functions:

- Medium for biochemical reactions
- Supports DNA replication and transcription
- Maintains internal organization

4.Nucleolus

- Dense, spherical, **non-membranous structure**
- Usually **1–4 nucleoli per nucleus**
- Rich in RNA and proteins

Functions:

- Synthesis of **ribosomal RNA (rRNA)**
- Assembly of **ribosomal subunits**
- Plays a role in protein synthesis indirectly

5. Chromatin

- Network of **DNA + histone proteins**
- Appears as thread-like structures

Types:

1. **Euchromatin**
    - Loosely packed
    - Transcriptionally active
2. **Heterochromatin**
    - Densely packed
    - Transcriptionally inactive

During cell division → chromatin condenses into **chromosomes**

Functions:

- Stores genetic information
- Controls heredity
- Regulates gene expression

FUNCTIONS OF THE NUCLEUS

1. **Control of Cellular Activities**

- Regulates metabolism, growth, differentiation
- Achieved by controlling **gene expressio**

2. **Storage and Transmission of Genetic Information**

- DNA contains genes responsible for traits
- Ensures **inheritance** from one generation to another

3. **DNA Replication**

- Occurs during **S-phase of cell cycle**
- Ensures each daughter cell gets identical DNA

4. **Transcription (RNA Synthesis)**

- DNA → mRNA, tRNA, rRNA
- mRNA carries instructions for protein synthesis
- **5. Regulation of Protein Synthesis**
- Nucleus controls which proteins are produced
- Nucleolus forms ribosomes

6. **Cell Division Control**

- Coordinates:
    - **Mitosis** (growth, repair)
    - **Meiosis** (gamete formation)

7. Cell Cycle Regulation

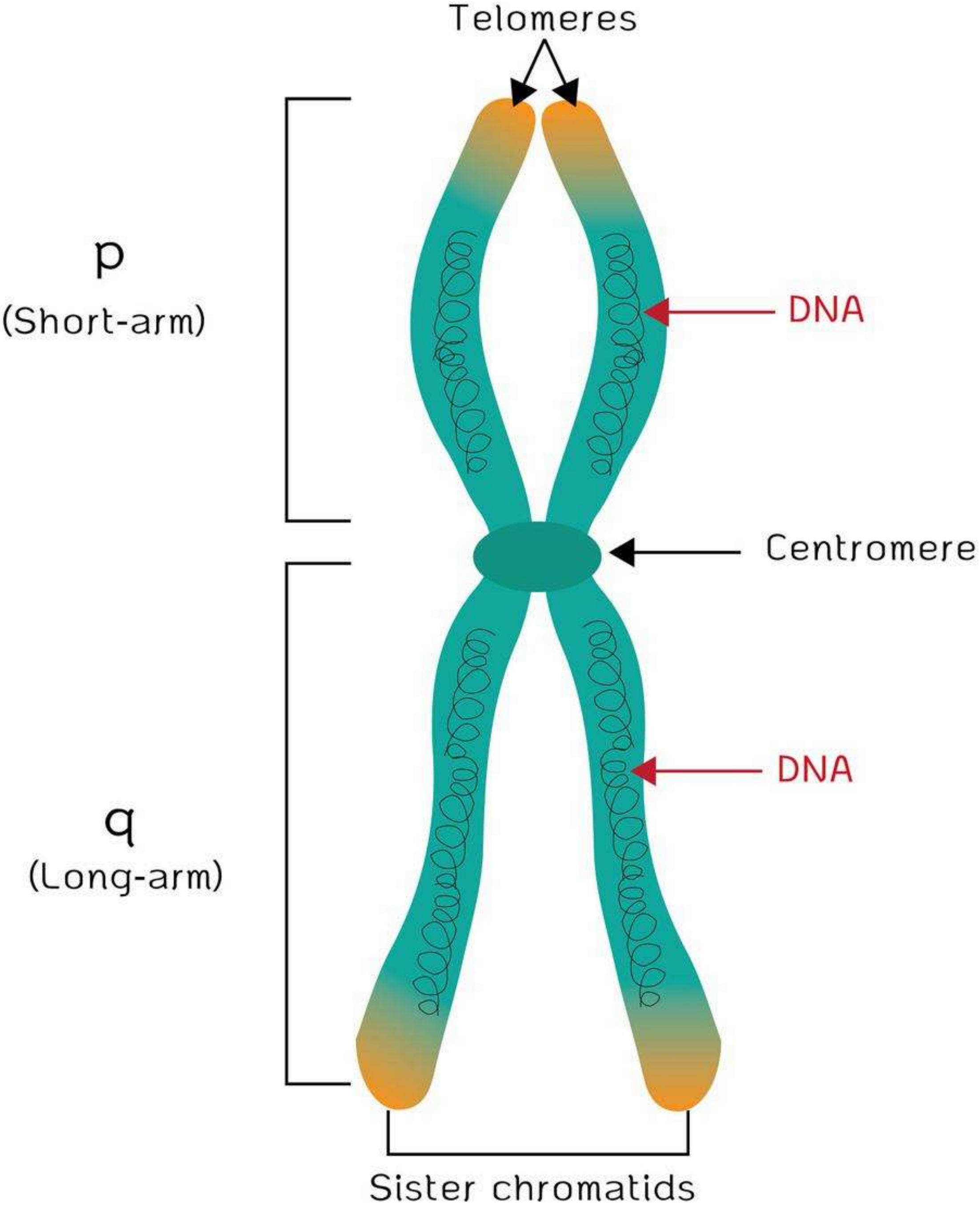
- Controls checkpoints

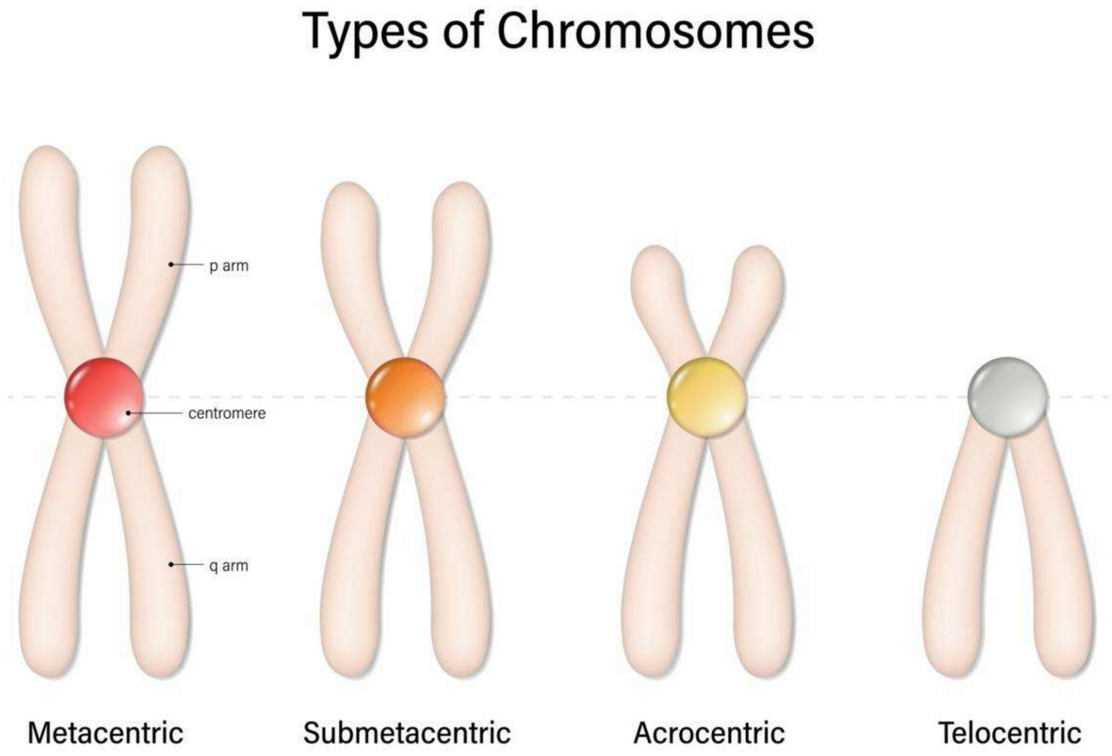
- Prevents abnormal division (e.g., cancer)

Chromosome

Labeled Diagram of Chromosome

CHROMOSOME STRUCTURE





1. INTRODUCTION

A **chromosome** is a highly organized, thread-like structure found in the nucleus of eukaryotic cells. It is composed of **DNA and proteins (mainly histones)** and carries hereditary information in the form of **genes**.

- In a non-dividing cell, DNA exists as **chromatin** (loosely coiled)
- During cell division, chromatin condenses into **visible chromosomes**

Thus, chromosomes are the **vehicles of heredity** and control all cellular functions.

2. DISCOVERY OF CHROMOSOMES

- First observed by **Karl Wilhelm von Nägeli (1842)** in plant cells
- The term **"chromosome"** (meaning *colored body*) was coined by **Heinrich Wilhelm Waldeyer (1888)**
- The role of chromosomes in heredity was explained by **Walter Sutton** and **Theodor Boveri**

This led to the **Chromosomal Theory of Inheritance**, stating that genes are located on chromosomes.

3. NUMBER OF CHROMOSOMES

- The chromosome number is **species-specific and constant**.
- **Types:**
- **Diploid (2n):** Body (somatic) cells
- **Haploid (n):** Gametes (sperm and egg)
- **Examples:**

Organism	**Chromosome Number**
Human	46 (2n)
Frog	26
Fruit fly	8
Rice	24

Fertilization restores diploid number (n + n = 2n)

4. SIZE OF CHROMOSOMES

- Length varies from **0.2 µm to 50 µm**
- Diameter ≈ **0.2–2 µm**
- Largest chromosomes → **amphibians**
- Smallest → **fungi**
- **Factors affecting size:**
- Amount of DNA
- Degree of coiling (condensation)

5. SHAPE OF CHROMOSOMES

Shape depends on the **position of the centromere**:

Types:

Metacentric

Centromere in middle

Arms equal → **V-shape**

Submetacentric

Slightly off-center

Arms unequal → **L-shape**

Acrocentric

Centromere near one end

One arm very short → **J-shape**

Telocentric

Centromere at end

Only one arm → **I-shape**

6. STRUCTURE OF CHROMOSOME (ULTRA-DETAIL)

- A chromosome at metaphase consists of the following parts:

1. Chromatids

- Two identical halves of a replicated chromosome
- Joined at the centromere
- Each contains a **single DNA double helix**

2. Centromere

- Primary constriction
- Divides chromosome into:
- **p-arm (short arm)**
- **q-arm (long arm)**

Essential for movement during cell division

3. Kinetochore

- Disc-shaped protein structure on centromere

- Attaches spindle fibers during mitosis/meiosis

4. Telomeres

- Terminal ends of chromosome
- Made of repetitive DNA sequences

Functions:

- Prevent DNA damage
- Maintain stability
- Prevent fusion of chromosomes

5. Secondary Constriction (NOR)

- Region responsible for nucleolus formation
- Contains genes for rRNA

6. Chromonema

- Coiled thread inside chromatid
- Shows bead-like structures

7. Chromomeres

- Bead-like structures on chromonema
- Represent condensed regions of DNA

8. Matrix

- Ground substance of chromosome
- Provides structural support

DNA Packaging (Important Concept)

- DNA is packed in multiple levels:
- DNA double helix
- Wrapped around histones → **Nucleosome**
- Coiled into chromatin fiber
- Supercoiled → Chromosome

This compact packaging allows long DNA to fit inside the nucleus.

7. FUNCTIONS OF CHROMOSOMES

1. Storage of Genetic Material

- Chromosomes carry **genes (DNA segments)**
- Determine traits like height, eye color, etc.

2. Transmission of Heredity

- Pass genetic information from parents to offspring

3. Role in Cell Division

- Ensure equal distribution of DNA
- Centromere and spindle fibers help movement

4. Regulation of Cellular Activities

- Control protein synthesis via gene expression

5. Protection of Genetic Material

- Telomeres prevent loss of important DNA

6. Variation and Evolution

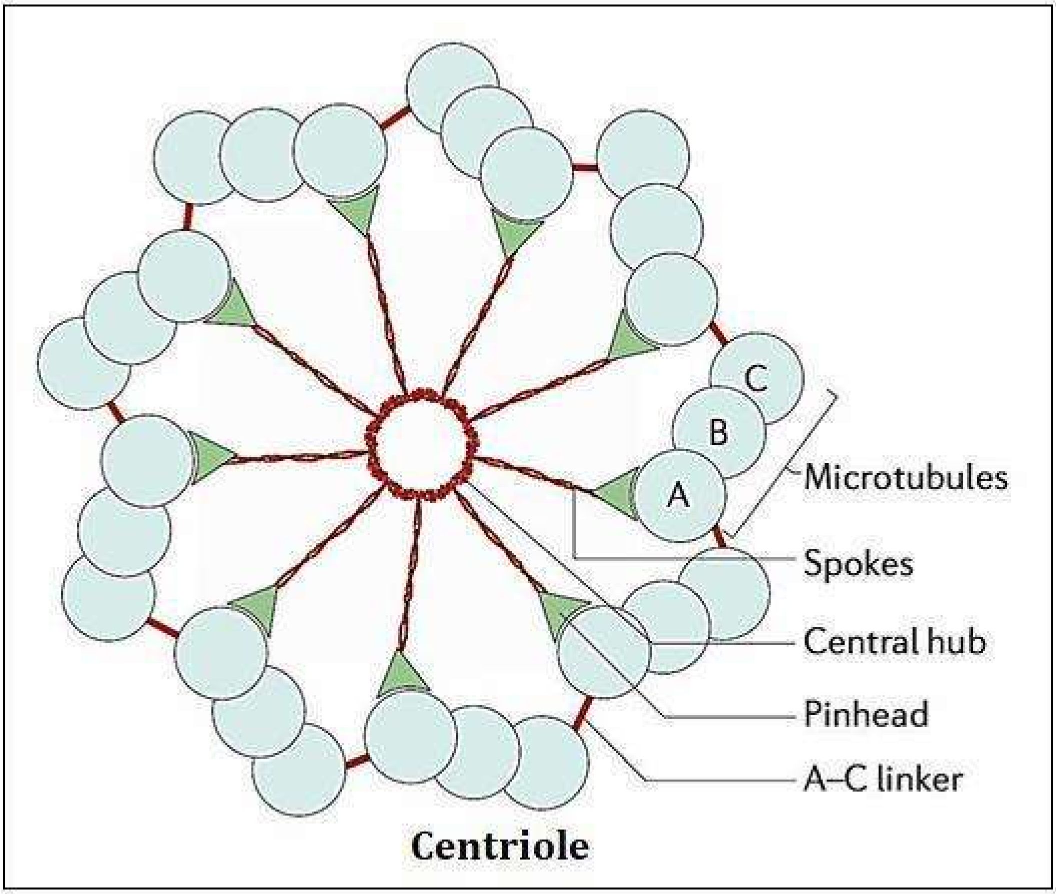
- Mutation and recombination create diversity
- Basis of evolution
- **QUICK SUMMARY TABLE**

Feature	**Description**
Composition	DNA + Histone proteins
Location	Nucleus
Visible stage	During cell division
Key parts	Chromatid, centromere, telomere

- Role
- Heredity + control of cell

Centrosome — Structure and Function

Diagram of Centrosome



1. INTRODUCTION

The **centrosome** is a **non-membranous organelle** found mainly in **animal cells**, located near the nucleus. It acts as the **microtubule organizing center (MTOC)** and plays a key role in **cell division**.

It consists of **two centrioles** arranged perpendicular to each other.

2. STRUCTURE OF CENTROSOME (DETAILED)

A. Centrosome

- Small, dense region in cytoplasm near nucleus
- Surrounded by a protein-rich matrix called **pericentriolar material (PCM)**

Function: Organizes microtubules and spindle fibers

B. Centrioles

Each centrosome contains **two centrioles**:

Key Features:

- Cylindrical structures
- Arranged at **right angles (90°)** to each other
- Each centriole has a **9+0 arrangement**

C. Ultrastructure of Centriole

- Made of **9 sets of microtubule triplets** (no central tubules)
- Each triplet has:
  - A-tubule (complete)
  - B and C tubules (incomplete)

This is called the **9+0 pattern**

D. Cartwheel Structure (in immature centrioles)

- Central hub with radial spokes
- Helps in centriole formation

E. Pericentriolar Material (PCM)

- Dense protein matrix surrounding centrioles
- Contains proteins like **tubulin**

Function: Nucleation (formation) of microtubules

3. FUNCTIONS OF CENTROSOME

1. **Spindle Formation During Cell Division**

- Forms **spindle fibers** in mitosis and meiosis
- Helps in **chromosome movement**

2. Microtubule Organization

- Acts as **MTOC (Microtubule Organizing Center)**
- Maintains cell shape and structure

3. Formation of Cilia and Flagella

- Centrioles form **basal bodies**
- Help in formation of:
  - Cilia (movement of mucus)
  - Flagella (e.g., sperm tail)

4. Cell Polarity and Organization

- Helps determine spatial arrangement of organelles

5. Cell Cycle Regulation

- Duplicates once per cell cycle
- Ensures proper division

CILIA AND FLAGELLA

1. INTRODUCTION

Cilia and **flagella** are **hair-like, motile appendages** present on the surface of certain eukaryotic cells. They help in **movement** of the cell or movement of substances over the cell surface.

Both have a similar internal structure but differ in **length, number, and function**

2. STRUCTURE OF CILIA AND FLAGELLA

Both cilia and flagella have the same basic structure:

A. Plasma Membrane

Outer covering

Continuous with cell membrane

- **B. Axoneme (Core Structure)**

  The most important part

  It shows a **“9 + 2 arrangement”**:

  9 doublet microtubules arranged in a circle

  2 central single microtubules

  Each doublet consists of:

  A-tubule (complete)

  B-tubule (incomplete)

  Associated Structures:

  Dynein arms → help in movement

  Radial spokes → provide stability

  Nexin links → connect doublets

This structure is responsible for **beating motion**

C. Basal Body

- Located at the base
- Similar to centriole (**9+0 arrangement**)

Function:

- Anchors cilia/flagella
- Initiates their formation

3. FUNCTIONS OF CILIA

- Move substances over cell surface e.g., mucus movement in respiratory tract
- Help in feeding in some organisms
- Create water currents (in protozoa)

4. FUNCTIONS OF FLAGELLA

- Help in **locomotion (movement of cell)**
  e.g., movement of sperm

- Propel the cell in a specific direction

5. DIFFERENCE BETWEEN CILIA AND FLAGELLA

Feature	Cilia	Flagella
Length	Short	Long
Number	Numerous	Few (1–4)
Movement	Coordinated, rhythmic	Wave-like
Function	Move substances over surface	Move entire cell

CELL INCLUSION

1. INTRODUCTION

Cell inclusions are **non-living, non-membranous substances** present in the cytoplasm of cells. They are not permanent cell structures but are formed as a result of **metabolic activities**.

They mainly serve as **storage materials, secretory products, or waste substances**.

2. CHARACTERISTICS OF CELL INCLUSIONS

- Non-living (inactive) components
- Usually **not bounded by membranes**
- May appear and disappear depending on cell activity
- Found in both plant and animal cells
- Do not perform vital metabolic functions directly

3. TYPES OF CELL INCLUSIONS

Cell inclusions can be broadly classified into the following:

A. Reserve (Storage) Materials

Stored food substances used when required

Examples:

- **Glycogen granules** (animal cells) → storage form of glucose

- **Starch grains** (plant cells) → carbohydrate reserve
- **Lipid droplets (fat)** → energy storage

Function: Provide energy and nutrients during need

B. Secretory Materials

Substances produced by cells for secretion

Examples:

- Enzymes
- Hormones
- Mucus

Function: Help in various physiological activities

C. Pigment Granules

Colored substances present in cells

Examples:

- **Melanin** → gives color to skin, hair
- **Carotenoids** → plant pigments
- **Hemoglobin** (in RBCs)

Function: Coloration and protection (e.g., UV protection)

D. Waste Products

Metabolic waste stored temporarily

Examples:

- Uric acid crystals
- Calcium oxalate crystals

Function: Waste storage before removal

E. Crystals (Common in Plants)

- Calcium carbonate
- Calcium oxalate (raphides, druses)

Function: Protection against herbivores and storag

4. FUNCTIONS OF CELL INCLUSIONS

1. **Storage of Food**
   - Store carbohydrates, fats, and proteins

2. Energy Reserve

- Provide energy during starvation

3. **Pigmentation**
   - Provide color to cells and tissues
4. **Protection**

   Crystals may protect against predators

5. Waste Management

- Store metabolic waste temporarily